

Assignment 2 – Optimizer Performance on Non-Convex Functions

1. Objective

The objective was to implement and compare five gradient-based optimization algorithms on non-convex functions and study their performance for different learning rates.

Optimizers: Gradient Descent, Gradient Descent with Momentum, Adam, RMSprop, and Adagrad.

Learning rates: 0.01, 0.05, and 0.1.

Functions: Rosenbrock Function and Simplified 1D Ackley Function.

2. Optimization

Optimization is the process of finding input values that minimize or maximize a function. In this assignment, the goal was minimization.

Gradient-based methods use the gradient to determine the direction of change. The basic update rule is:

$$\mathbf{x}(\text{new}) = \mathbf{x}(\text{old}) - \alpha \times \text{gradient}$$

Here, α is the learning rate. The parameter is updated in the opposite direction of the gradient.

3. Functions Used

Rosenbrock Function:

$$f(x,y) = (1-x)^2 + 100(y-x^2)^2$$

Its global minimum is at (1,1), where $f(x,y) = 0$. It has a narrow curved valley, which makes optimization challenging.

Simplified 1D Ackley Function:

The function has an oscillating landscape with multiple local minima. Its global minimum is near $x = 0$, where $f(x) = 0$.

4. Why Non-Convex Functions?

Non-convex functions can contain local minima, flat regions, narrow valleys, and oscillations. They are therefore useful for comparing the behaviour of different optimization algorithms.

5. Optimizers

Optimizer	Main Idea
Gradient Descent	Uses the current gradient to update parameters.
Momentum	Uses previous update information to improve movement.
RMSprop	Adapts the learning rate using squared gradients.
Adagrad	Uses accumulated squared gradients; learning rate decreases over time.
Adam	Combines momentum and adaptive learning rates.

6. Learning Rate

The learning rate controls the size of each optimization step. A small learning rate is usually more stable but may converge slowly. A large learning rate can be faster but may cause overshooting, oscillation, or divergence.

7. Experimental Setup

Each of the five optimizers was tested with three learning rates on both functions.

5 optimizers $\times$ 3 learning rates = 15 experiments per function

2 functions $\times$ 15 experiments = **30 total experiments**

For each experiment, the final point, function value, number of iterations, execution time, and convergence status were recorded.

8. Convergence Criteria

The optimization process stops when a stopping condition is satisfied. The assignment used a maximum of 10,000 iterations along with gradient, function-value, and step tolerances.

9. Performance Comparison

The optimizers were compared using final function value, number of iterations, execution time, stability, and sensitivity to the learning rate. A lower function value indicates a solution closer to the minimum.

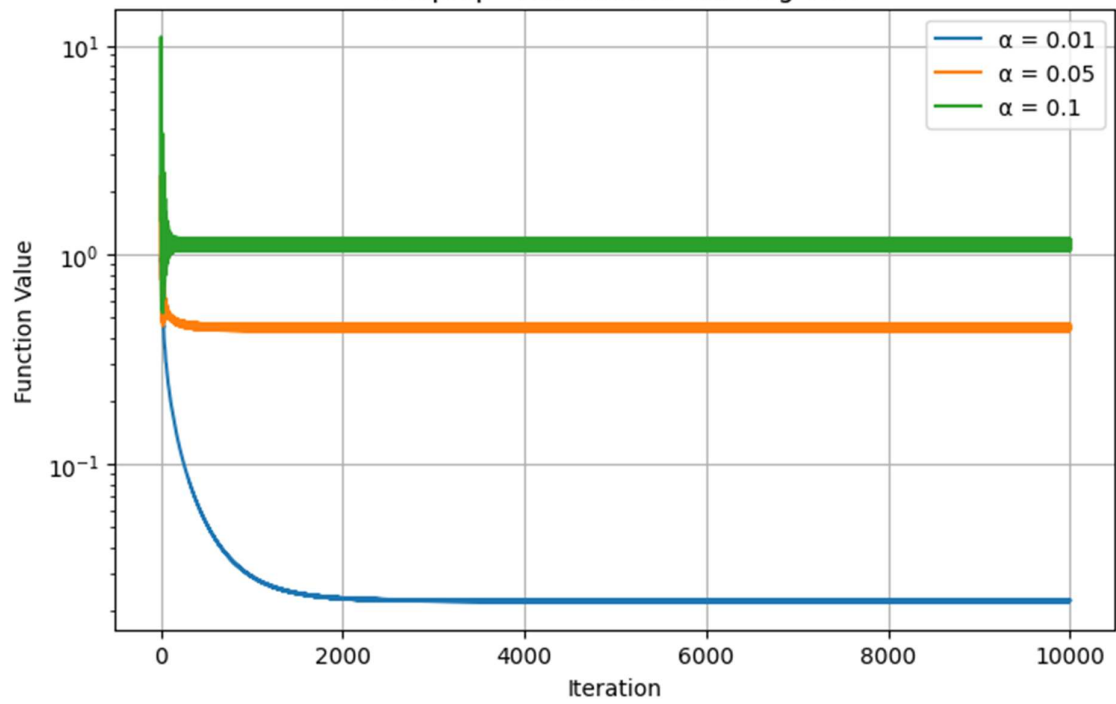
10. Methodology

1. Define the Rosenbrock function and its gradient.
2. Define the simplified Ackley function and its gradient.
3. Implement all five optimizers from scratch.
4. Test learning rates 0.01, 0.05, and 0.1.
5. Run the optimizers on both functions.
6. Record performance metrics and convergence status.
7. Generate convergence plots and summary tables.
8. Export the results to Excel and compare the optimizers.

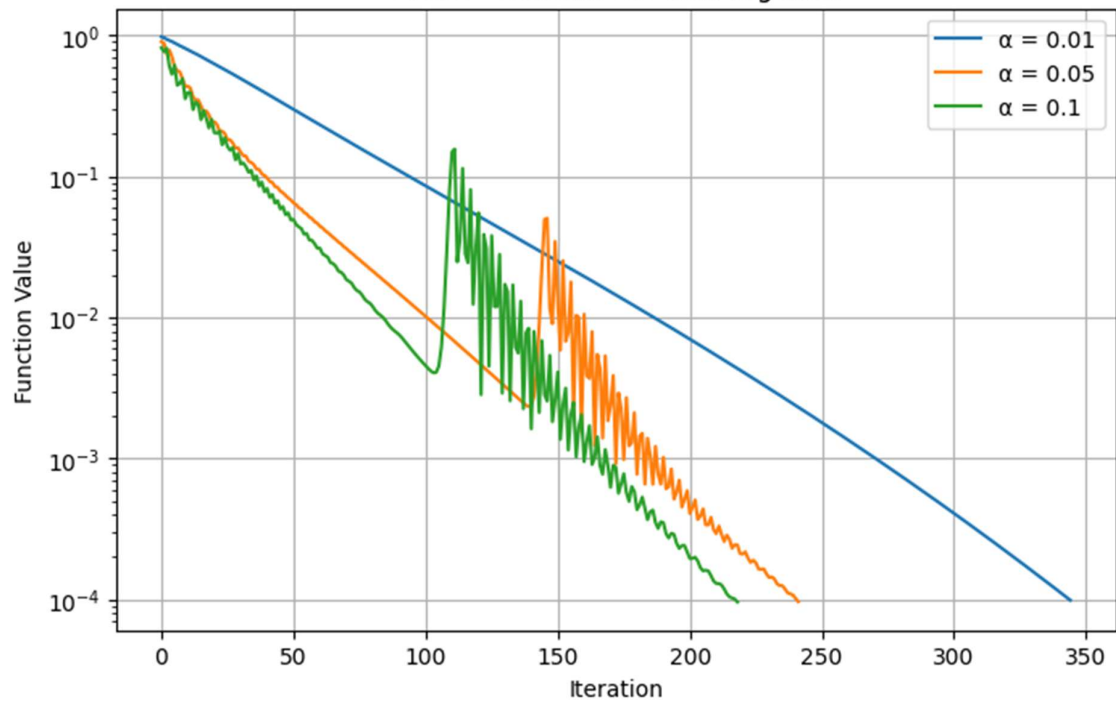
11. Conclusion

The experiment demonstrated the behaviour of different optimization algorithms on non-convex functions. Gradient Descent provides a simple foundation but is sensitive to the learning rate. Momentum improves movement using previous updates, while RMSprop and Adagrad adapt the learning rate. Adam combines momentum and adaptive learning rates and can provide fast and stable optimization. Overall, optimizer performance depends on the function, learning rate, initial point, and convergence criteria.

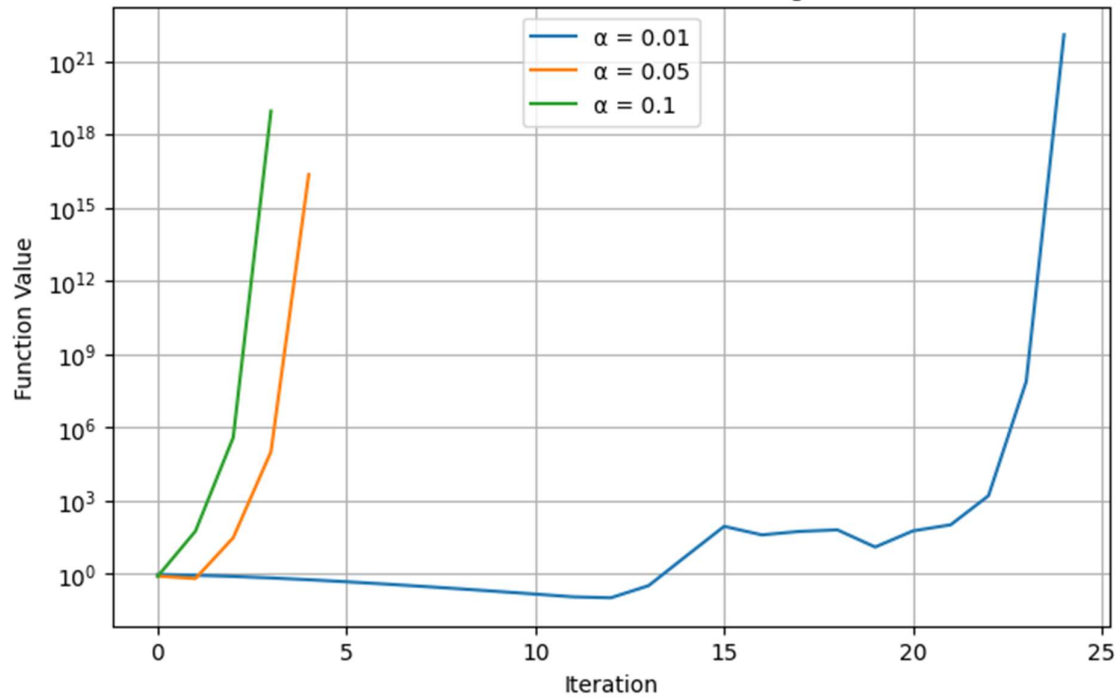
RMSprop - Rosenbrock Convergence



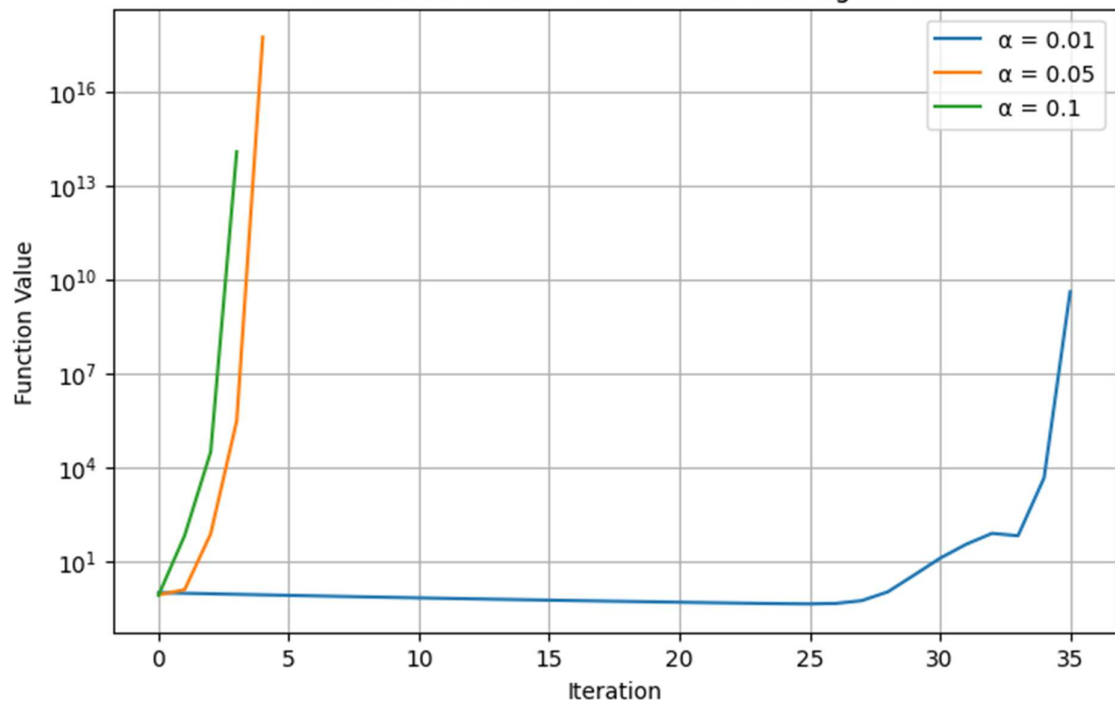
Adam - Rosenbrock Convergence



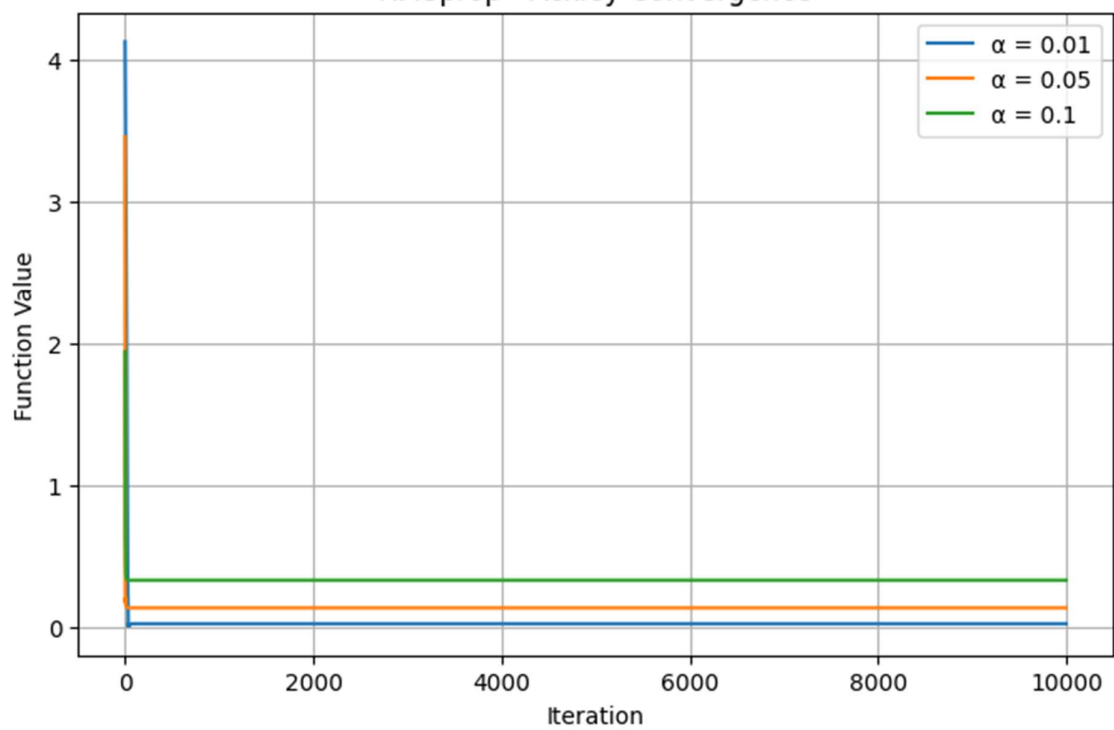
Momentum - Rosenbrock Convergence



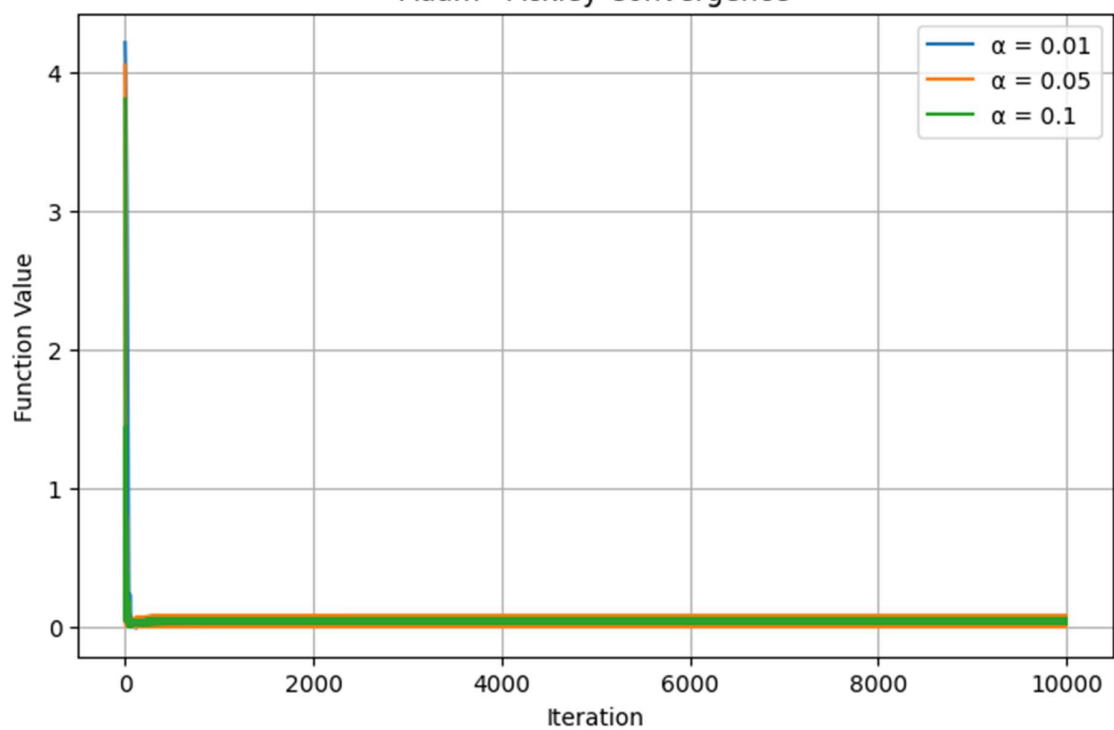
Gradient Descent - Rosenbrock Convergence

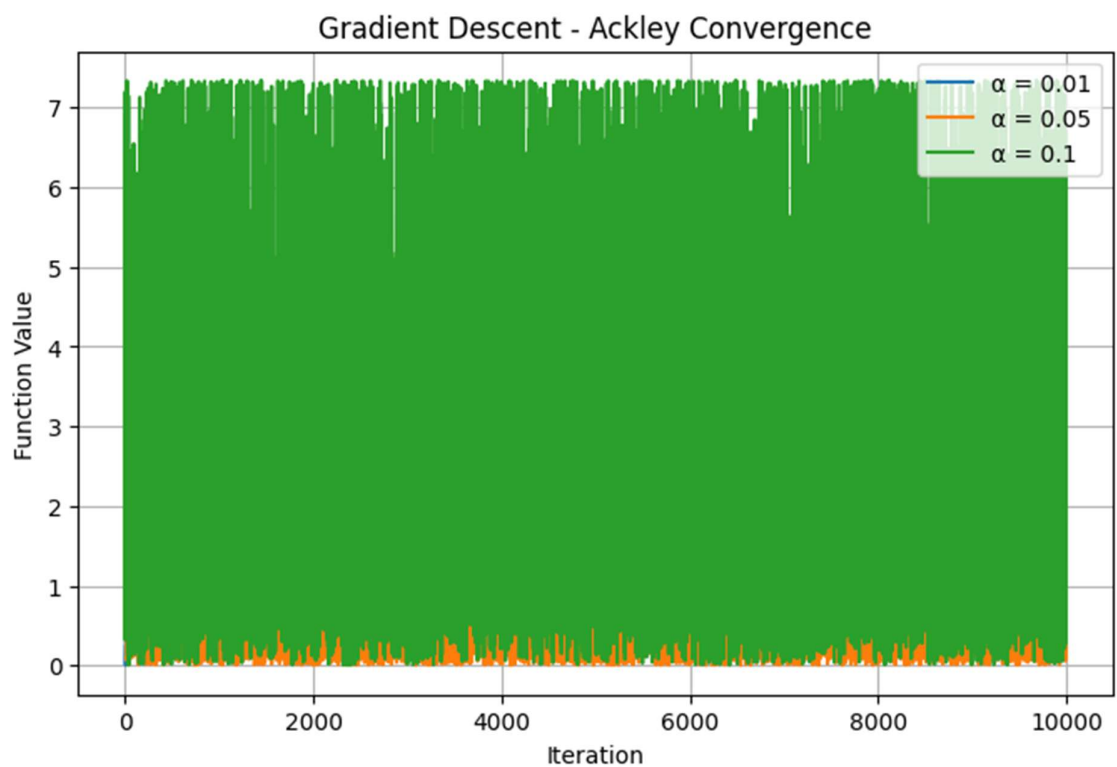
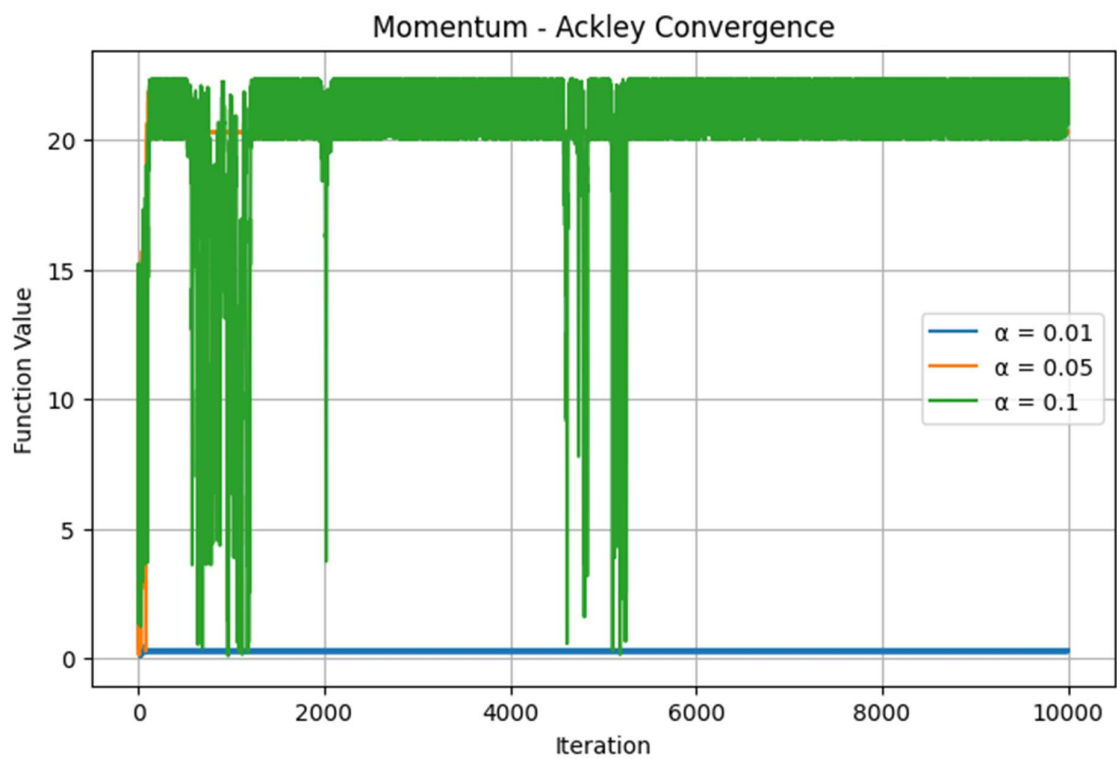


RMSprop - Ackley Convergence



Adam - Ackley Convergence





Ackley: Optimizer Comparison ($\alpha = 0.01$)

